# Database Technologies – Assignment 10

## PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle,
Perimeter = 2*(L+W) and Area = L*W.

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE sp_perimeter_area(p_length INT, p_width INT)
    ->
    -> BEGIN
    ->
    -> DECLARE perimeter INT;
    -> DECLARE area INT;
    ->
    -> SET perimeter = 2*(p_length + p_width);
    -> SET area = p_length * p_width;
    ->
    -> SELECT p_length AS Length, p_width AS Width, perimeter, area;
    ->
    -> END;
    ->
    -> $$
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL sp_perimeter_area(10, 20);
+--------+-------+-----------+------+
| Length | Width | perimeter | area |
+--------+-------+-----------+------+
|     10 |    20 |        60 |  200 |
+--------+-------+-----------+------+
1 row in set (0.002 sec)

Query OK, 0 rows affected (0.002 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
mysql> CREATE PROCEDURE sp_square_cubes(num INT)
    ->
    -> BEGIN
    ->
    -> DECLARE square INT;
    -> DECLARE cubes INT;
    ->
    -> SET square = num*num;
    -> SET cubes = num*num*num;
    ->
    -> SELECT num AS NUMBER, square, cubes;
    ->
    -> END;
    ->
    -> $$
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL sp_square_cubes(7)
    -> CALL sp_square_cubes(7);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to yo
ur MySQL server version for the right syntax to use near 'CALL sp_square_cubes(7)' at line 2
mysql> CALL sp_square_cubes(7);
+--------+--------+-------+
| NUMBER | square | cubes |
+--------+--------+-------+
|      7 |     49 |   343 |
+--------+--------+-------+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.001 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are
C= (F-32) *5/9 F= 9/5*C + 32

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE cnvrt_cf(num float, ch char)
    -> BEGIN
    ->     DECLARE res float;
    ->     IF ch = 'f' THEN
    ->         SET res = (num - 32) * 5/9;
    ->         SELECT res AS Celsius;
    ->     ELSE
    ->         SET res = (9/5) * num + 32;
    ->         SELECT res AS Fahrenheit;
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL cnvrt_cf(40, 'c')
    -> CALL cnvrt_cf(40, 'c');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to yo
ur MySQL server version for the right syntax to use near 'CALL cnvrt_cf(40, 'c')' at line 2
mysql> CALL cnvrt_cf(40, 'c');
+------------+
| Fahrenheit |
+------------+
|        104 |
+------------+
1 row in set (0.002 sec)

Query OK, 0 rows affected (0.002 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE cnvrt_yrd_ft(num INT)
    ->
    -> BEGIN
    ->
    -> DECLARE yard INT;
    -> DECLARE feet INT;
    -> DECLARE inch INT;
    ->
    -> SET yard = num / 36;
    -> SET feet = (num % 36) / 12;
    -> SET inch = (num % 36) % 12;
    ->
    -> SELECT num, yard, feet, inch;
    ->
    -> END;
    ->
    -> $$
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL cnvrt_yrd_ft(124);
+------+------+------+------+
| num  | yard | feet | inch |
+------+------+------+------+
|  124 |    3 |    1 |    4 |
+------+------+------+------+
1 row in set (0.002 sec)

Query OK, 0 rows affected (0.002 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE real_num(num1 FLOAT, num2 FLOAT)
    -> BEGIN
    ->
    ->     DECLARE product FLOAT;
    ->     SET product = num1 * num2;
    ->
    ->     IF product >= 100 THEN
    ->         SELECT product AS result, 'GREATER' AS MESSAGE;
    ->     ELSE
    ->         SELECT product AS result, 'LOWER' AS MESSAGE;
    ->     END IF;
    ->
    -> END $$
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL real_num(20, 6);
+--------+---------+
| result | MESSAGE |
+--------+---------+
|    120 | GREATER |
+--------+---------+
1 row in set (0.002 sec)

Query OK, 0 rows affected (0.002 sec)
```